Monetary Policy Uncertainty and the Dispersion of Inflation Expectations in South Africa

Research Proposal

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Motivation

Since the Global Financial Crisis, uncertainty has re-emerged as a first-order driver of macroeconomic outcomes, affecting investment, hiring, and consumption decisions (Bloom 2009). A growing strand of this literature focuses specifically on *monetary policy uncertainty* (MPU) – uncertainty about the future stance and reaction function of central banks – and its effect on the formation of inflation expectations (Arce-Alfaro and Blagov 2023). Where the private sector is less certain about the monetary authority's future actions, it should also be less certain, and more disperse, in its inflation forecasts. For a small, open, inflation-targeting economy such as South Africa, this channel is compounded by exchange-rate uncertainty and spillovers from US monetary policy uncertainty, both of which have been shown to matter for domestic volatility and expectations formation (Mumtaz and Zanetti 2013; Bianchi, Kung, and Tirskikh 2023).

Despite this, little empirical work has examined how MPU – domestic and foreign – relates to the *cross-sectional dispersion* of inflation expectations across different types of economic agents in South Africa. Disagreement (dispersion) is itself informative. It proxies the confidence agents place in their own forecasts and, unlike point forecasts, captures how uncertainty is transmitted unevenly across the economy.

Research Questions

1. Does monetary policy uncertainty – in South Africa and in the United States – raise the dispersion of inflation expectations among South African businesses, financial analysts, and labour representatives?

2. Does exchange-rate uncertainty (USD/ZAR volatility) amplify or operate independently of this relationship?
3. Does the sensitivity of expectations dispersion to MPU vary by forecast horizon (current year through five years ahead) or by respondent sector?

Data

The analysis draws on quarterly data assembled from the following sources:

Series	Source	Frequency
Dispersion (SD) of inflation expectations, by sector (business, finance, labour) and horizon	BER Inflation Expectations Survey	Quarterly
South African monetary policy uncertainty	Calculated market-based measure	Daily, aggregated
Alternative South African monetary policy uncertainty	Codera news-based measure	Daily, aggregated
US monetary policy uncertainty	Baker, Bloom & Davis EPU – monetary policy category	Monthly, aggregated
USD/ZAR exchange-rate uncertainty	Own GARCH(1,1) conditional volatility, USD/ZAR daily returns	Daily, aggregated

All series are harmonised to a common quarterly frequency, spanning approximately 1985 (uncertainty series) to 2026, with the dispersion measures available from 2011 onward.

Proposed Methodology

1. **Uncertainty measurement:** estimate a GARCH(1,1) model on USD/ZAR log returns to derive a time-varying, model-based exchange-rate uncertainty proxy.
2. **Empirical model:** regress each sector-horizon dispersion measure on lagged domestic and US MPU, exchange-rate volatility, and relevant macro controls (e.g. inflation, policy rate, output gap), using HAC-robust standard errors to address serial correlation typical of overlapping quarterly survey data. We estimate a proxy Baysian SVAR - with SA MPU, US MPU and USDZAR volatility as external instruments
3. **Robustness:** assess sensitivity to horizon and sector, and to alternative uncertainty proxies (news based measures); check series for stationarity and structural breaks given the long sample period (repo-era transition, 2008 GFC, 2020 pandemic).

Contribution

This paper contributes to the literature in four ways. First, it extends the monetary policy uncertainty–expectations literature (Mumtaz and Zanetti 2013; Arce-Alfaro and Blagov 2023) to an emerging-market, inflation-targeting setting where MPU is compounded by exchange-rate risk and external spillovers – a combination that has not been jointly estimated even though both channels are independently well documented for South Africa. Second, most existing work on expectations dispersion relies on published group averages; because this project uses BER respondent-level microdata for the business panel, it can construct genuine within-group cross-sectional dispersion rather than inferring disagreement from the spread of sector means – a stronger and more direct proxy for disagreement. Third, by decomposing dispersion responses by both respondent sector (business, finance, labour) and forecast horizon (current year to five years ahead), the paper can speak to whether MPU erodes near-term forecast precision, long-run anchoring, or both – a distinction the existing literature (largely focused on single-horizon or point-forecast measures) does not resolve. Fourth, on methodology, using GARCH-based domestic policy-rate and exchange-rate uncertainty proxies alongside a news-based MPU series lets the paper cross-check a model-based and a perception-based measure of the same underlying uncertainty, addressing a known fragility in this literature (Bianchi, Kung, and Tirskikh 2023) where results can be sensitive to how uncertainty itself is measured.

References

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